

BUFFALO, SD, USA

WMO: 726627

Lat: 45.604N

Lon: 103.546W

Elev: 3004

StdP: 13.17

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 94037

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -12.4	(c) -6.8	(d) -20.1	(e) 2.0	(f) -11.8	(g) -13.9	(h) 2.9	(i) -6.2	(j) 42.0	(k) 30.6	(l) 39.0	(m) 26.5	(n) 11.4	(o) 300	(p) 0.736

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 26.4	(c) 95.6	(d) 65.6	(e) 91.5	(f) 65.0	(g) 88.2	(h) 64.5	(i) 70.5	(j) 85.6	(k) 68.7	(l) 84.2	(m) 67.0	(n) 82.3	(o) 12.4	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.0	107.1	76.7	63.8	99.2	74.6	62.0	92.9	73.4	36.3	85.8	34.7	83.7	33.2	82.1	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
36.5	32.1	27.5	-20.0	102.0	4.3	4.4	-23.0	105.2	-25.5	107.8	-27.9	110.3	-31.0	113.5
			-20.3	73.4	4.2	2.0	-23.3	74.8	-25.8	76.0	-28.1	77.1	-31.1	78.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.6	21.0	21.5	33.4	43.6	53.6	64.5	73.0	70.3	61.0	46.3	34.0	23.6
	DBStd	21.20	14.27	14.73	14.11	9.93	8.74	7.25	6.33	6.79	9.15	9.74	11.74	13.44
	HDD50	4032	898	800	529	234	62	3	0	0	15	187	486	818
	HDD65	7678	1363	1218	981	642	365	94	10	28	181	584	929	1283
	CDD50	2433	0	1	13	42	174	437	713	629	346	71	7	0
	CDD65	604	0	0	0	1	11	78	258	192	61	3	0	0
	CDH74	7860	0	0	7	45	189	1026	3122	2407	978	86	0	0
	CDH80	3509	0	0	0	8	39	366	1536	1129	409	22	0	0
	WSAvg	12.4	12.8	12.2	13.1	13.7	13.5	12.3	10.9	11.0	11.6	12.8	12.3	12.2
	PrecAvg	15.1	0.4	0.4	0.7	1.5	2.6	3.0	2.0	1.5	1.1	1.3	0.5	0.3
Precipitation	PrecMax	20.4	1.6	1.1	2.1	3.5	6.4	5.4	5.7	3.6	3.0	4.5	3.4	1.0
	PrecMin	8.0	0.0	0.0	0.1	0.2	0.5	1.3	0.3	0.2	0.1	0.1	0.0	0.0
	PrecStd	3.3	0.4	0.2	0.5	0.7	1.6	1.1	1.2	0.9	0.7	1.1	0.6	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.1	62.6	73.4	79.6	84.6	94.9	102.1	99.5	94.8	82.5	69.1	56.9
		MCWB	42.1	45.7	50.3	54.3	61.4	66.5	67.0	67.0	63.8	57.6	49.5	41.5
	2%	DB	47.3	52.3	65.7	72.1	79.3	88.3	97.1	94.6	89.8	74.8	60.9	49.9
		MCWB	38.5	40.9	47.4	51.9	60.1	65.3	66.8	64.7	62.2	54.4	45.7	38.7
	5%	DB	42.7	45.0	59.5	66.1	74.8	83.8	92.5	90.6	84.7	68.2	55.0	45.1
		MCWB	35.8	37.2	44.4	49.3	57.5	64.5	66.0	64.7	59.8	50.9	42.6	36.2
	10%	DB	39.0	40.2	53.3	61.2	70.3	80.0	88.7	86.7	79.5	62.6	50.1	40.4
		MCWB	33.6	34.3	42.0	46.8	55.7	62.6	65.8	64.0	58.4	48.8	40.3	33.9
	0.4%	WB	42.9	46.1	52.0	56.1	65.9	71.9	73.4	71.3	67.9	59.2	50.2	43.4
		MCDB	53.2	60.5	69.8	75.5	79.6	83.7	85.6	88.1	84.8	77.7	67.3	53.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	38.9	41.2	48.3	53.1	62.0	69.0	71.2	69.0	64.4	55.7	46.4	39.6
		MCDB	46.3	51.1	63.2	69.9	73.9	81.8	86.4	85.7	82.7	70.7	59.4	47.8
	5%	WB	36.1	37.6	45.5	50.4	59.3	66.5	69.5	67.4	62.2	52.6	43.4	36.7
		MCDB	42.0	44.5	59.0	64.7	72.1	79.7	85.4	83.7	80.2	65.7	54.5	44.4
	10%	WB	33.8	34.6	42.5	47.9	56.7	64.4	67.7	65.7	59.8	49.7	40.5	34.0
		MCDB	38.8	39.7	52.7	59.9	68.5	77.8	83.3	81.2	75.8	61.4	49.2	40.3
	5% DB	MDBR	18.5	19.1	21.4	22.4	21.6	23.5	26.4	26.9	26.1	22.2	20.7	17.4
		MCDBR	21.3	24.6	31.7	32.3	28.8	29.9	32.5	32.7	34.3	32.0	27.7	21.4
		MCWBR	14.4	15.2	16.8	16.4	13.7	13.0	10.7	11.6	12.7	15.4	15.5	13.8
		MCDBR	20.8	23.3	30.1	30.3	26.2	27.0	27.7	27.6	30.6	29.7	27.2	21.3
	5% WB	MCWBR	14.6	15.0	16.5	16.1	13.8	13.2	11.3	11.5	12.2	14.8	15.7	14.4
Clear-Sky Solar Irradiance	taub		0.252	0.257	0.276	0.311	0.330	0.344	0.360	0.359	0.324	0.289	0.261	0.246
	taud		2.546	2.547	2.551	2.443	2.461	2.452	2.434	2.412	2.500	2.566	2.543	2.546
	Ebn at Noon		282	301	307	301	297	292	286	282	285	284	275	272
	Edh at Noon		20	24	27	34	34	35	35	34	29	23	19	17
All-Sky Solar Radiation	RadAvg		567	885	1239	1588	1869	2146	2232	1880	1444	937	614	460
	RadStd		36	65	99	91	188	158	94	102	93	93	59	25

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page